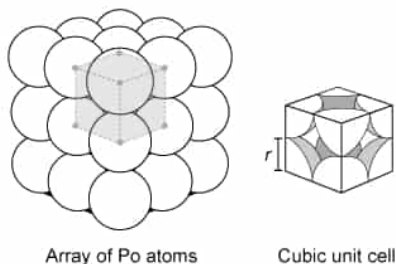


Atoms in a sample of solid polonium are packed in a repeating array that can be represented by a cubic unit cell, which is the simplest three-dimensional cross-section that repeats throughout the solid.



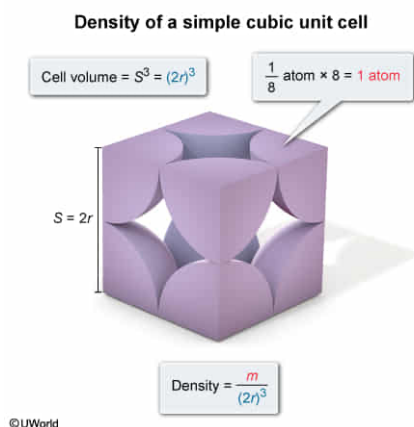
If the atomic radius r is 1.67×10^{-8} cm and the mass of one polonium atom is 3.47×10^{-22} g, which expression correctly calculates the density of a pure sample of solid polonium?

- ☐ A. $\frac{(3.47 \times 10^{-22} \text{ g})}{(1.67 \times 10^{-8} \text{ cm})^3}$ [17%]
- ☐ B. $\frac{8 \times (3.47 \times 10^{-22} \text{ g})}{(1.67 \times 10^{-8} \text{ cm})^3}$ [7%]
- ☒ C. $\frac{(3.47 \times 10^{-22} \text{ g})}{(2 \times 1.67 \times 10^{-8} \text{ cm})^3}$ [45%]
- ☐ D. $\frac{8 \times (3.47 \times 10^{-22} \text{ g})}{\left(\frac{4}{3}\pi\right)(1.67 \times 10^{-8} \text{ cm})^3}$ [28%]

Correct

44% answered correctly

Explanation:



The **density ρ** of a substance is defined as the **mass m** it contains per unit of **volume V** that it occupies. This physical property is expressed mathematically as:

$$\rho = \frac{m}{V}$$

Because materials tend to expand when heated and contract when cooled, the volume that a substance occupies can undergo minute changes with temperature. As a result, the exact density of a substance **varies slightly with temperature**.

On the most fundamental level, the density of a substance is the result of its atomic packing. Atoms that pack together more tightly achieve a greater density than molecules with a less space-efficient packing arrangement. Accordingly, the **density of an element** can be calculated based on its **atomic radius** and the **atomic packing** within its **unit cell** (ie, the density of the unit cell is equal to the density of a solid with a matching atomic arrangement).

Polonium atoms pack together in an arrangement that gives a simple cubic unit cell. As such, the length of one side S of the unit cell is equal to twice the atomic radius ($2r$). When this length is used, the **volume of the cube** (unit cell) is:

$$V = S^3 = (2r)^3$$

Substituting this volume expression into the density equation gives:

$$\rho = \frac{m}{(2r)^3}$$

Because each of the eight corners of the unit cell hold one-eighth of a polonium atom, the unit cell contains the equivalent mass of one polonium atom. Substituting the values for mass and the atomic radius into the equation above gives the density:

$$\rho = \frac{(3.47 \times 10^{-22} \text{ g})}{(2 \times 1.67 \times 10^{-8} \text{ cm})^3} = 9.31 \text{ g/cm}^3$$

(Choice A) The length of one side S of the unit cell is equal to *twice* the atomic radius ($2r$).

(Choice B) Although the unit cell cuts through eight atoms, the interior of the unit cell contains the equivalent mass of only one atom (one-eighth of each of the eight atoms).

(Choice D) The [volume of a sphere](#) is found using $4\pi r^3/3$, but this does not account for the volume between the spheres. The volume occupied by a solid sample contains both atoms and empty spaces between the atoms. As such, the volume of the unit cell cube must be used to calculate the density of the sample.

Educational objective:

The density of a substance varies slightly with temperature and is defined as the mass of the substance per unit of volume that it occupies. The density of an element can be calculated based on its atomic radius and the atomic packing within its unit cell.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402618
System:
Atomic Theory and Chemical Composition